

Research Article

Protective Effects of *Scrophularia striata* Extract on Testicular Function and Spermatogenesis in Cadmium-Exposed Mice: Role of Nitro-Oxidative Stress Modulation

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Received 19 May 2024; Accepted 16 December 2024

Academic Editor: Raul Sanchez

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Cadmium (Cd), an environmental pollutant present in soil, water, and food, is known to pose significant risks to male reproductive health by inducing oxidative stress and toxicity. *Scrophularia striata* extract (SSE) is rich in natural antioxidants that play a crucial role in neutralizing toxins and protecting various organs. This study aimed to evaluate the therapeutic potential of SSE in mitigating testicular damage and impairments in spermatogenesis induced by Cd using an animal model. Adult mice were randomly assigned to six groups: control, Cd, SSE 100 mg/kg, SSE 200 mg/kg, SSE 100 mg/kg plus Cd, and SSE 200 mg/kg plus Cd. Mice were intraperitoneally injected with 0.35 mg/kg of Cd for 35 days, with or without SSE. At the study's conclusion, various parameters were assessed, including sperm count, motility, viability, morphology, DNA integrity, testosterone levels, testicular histopathology, and the levels of malondialdehyde (MDA), nitric oxide (NO), and total antioxidant capacity (TAC) in the testes. The results demonstrated that SSE significantly improved spermatogenesis, as indicated by improved sperm count, motility, viability, morphology, and chromatin integrity. Additionally, testosterone levels and testicular histopathology were notably improved in SSE-treated groups. While Cd exposure significantly elevated oxidative stress in the testes, SSE treatment effectively enhanced antioxidant defense by increasing TAC levels and reducing NO and MDA levels. Overall, SSE provided substantial protection against testicular damage induced by chronic Cd exposure by modulating oxidative stress, highlighting its potential as a promising herbal therapy for addressing reproductive toxicity and male infertility linked to environmental pollutants.

Keywords: antioxidant; cadmium; oxidative stress; *Scrophularia striata*; sperm quality; testis

1. Introduction

Over the last few decades, men across various nations have undergone a significant decrease exceeding 50% in both the total number and concentration of sperm [1, 2]. This decline

has been implicated in around 15% of couples encountering difficulties conceiving despite engaging in regular sexual activity [1, 3]. Lately, there's been growing concern about the adverse effects of environmental pollution on the reproductive system, with many experts identifying it as a

significant contributor to the diminishing fertility rates among males [4].

Cadmium (Cd) is a prevalent environmental toxin found naturally, exposing millions worldwide to chronic Cd through various means, both in workplaces and everyday environments [3, 5]. Its extensive presence in nature and diverse applications in daily life make exposure inevitable [3, 6, 7]. Notably, cigarette smoking and the agricultural use of phosphate fertilizers containing Cd are significant sources of exposure, further exacerbating its entry into the food chain [6, 8]. Even at low doses, Cd can cause harmful effects, compounded by its long biological half-life of 20–40 years, leading to its accumulation in human tissues, particularly the kidneys, bones, and reproductive organs [9, 10]. Cd exposure triggers the production of reactive oxygen species (ROS), contributing to oxidative stress, inflammation, cellular damage, altered gene expression, and various health issues in humans and animals [11, 12]. Reproductive toxicity, particularly testicular injury, is a primary concern in Cd poisoning [12–14]. Research indicates that Cd levels in the blood rise with tobacco smoking, intensifying oxidative stress and diminishing antioxidant defenses in the semen of infertile males [13, 15]. Cd disrupts the blood–testis barrier, leading to germ cell apoptosis, testicular damage, dysfunction in Sertoli and Leydig cells, hormonal imbalance, erectile dysfunction, impaired spermatogenesis, reduced semen quality, and ultimately infertility [13, 16]. Experimental studies also highlight Cd's interference with spermatogenesis and its damaging effects on sperm nuclear DNA, indicating areas for further investigation [2, 13].

Antioxidants and the antioxidant defense systems are vital for protecting male reproductive organs against oxidative stress [17–19]. Oxidative stress stands as a primary contributor to tissue damage caused by Cd [6, 13]. Given the testis' abundance of polyunsaturated fatty acids and limited antioxidant capacity, it becomes a key target organ for oxidative stress, leading to diminished sperm viability, motility, and subsequent infertility [6, 20, 21]. Given the inevitability of Cd exposure and its impact on the testis, numerous studies have concentrated on identifying the specific molecular targets of Cd toxicity. Subsequently, the aim of the current research has been to discover substances that could mitigate or eradicate its harmful effects, paving the way for novel therapeutic strategies.

For a considerable period, natural remedies, particularly medicinal plants, served as the foundation and sometimes the sole treatment for illnesses, with their raw materials being utilized in the pharmaceutical industry [22]. Based on statistics from the World Health Organization, around 80% of the global population favors herbal extracts or their active ingredients for primary healthcare [23, 24]. Plants and their compounds hold promise as alternatives to chemical drugs, often with fewer side effects [23, 25, 26]. Researchers have delved into the utilization of antioxidants found in medicinal plants to combat toxicity induced by heavy metals like lead, Cd, and mercury in biological systems [27–29]. Recent investigations have honed in on plant-derived products with antioxidant

properties, aiming to protect cells and tissues from the adverse effects of Cd toxicity in both animals and humans [27, 30, 31].

Scrophularia striata (*S. striata*), commonly referred to as “Tashaneh Dari,” belongs to the Scrophulariaceae family and holds significant importance as a traditional herbal remedy in western Iran, where it has been utilized for centuries [32–34]. Residents of these areas regard it as a “miraculous plant” due to its historical use in treating burns, wounds, scabies, ulcers, hemorrhoids, eczema, and infectious diseases [35, 36]. Numerous studies have outlined the antioxidant, antimicrobial, antianxiety, antiproliferative, anti-inflammatory, neuroprotective, analgesic, antidepressant, and wound-healing properties of this plant [24, 35, 37]. The phytochemical examination of *S. striata* extract (SSE) revealed the existence of bioactive compounds, including iridoids (such as scropolioside A and scrovalentinolide), phenylpropanoids, terpenoids, phenolic acids (like cinnamic acid), flavonoids (such as quercetin, isorhamnetin-3-O-rutinoside, and nepitrin), and a single phenylpropanoid glycoside (acteoside 1) [34–37]. Recent studies have documented the antioxidant and anti-inflammatory capabilities of SSE, yet there has been no study conducted to assess the protective effects of SSE against reproductive toxicity.

Thus, the study aims to evaluate the effects of the hydroalcoholic extract of SSE on various parameters in mice with Cd-induced reproductive toxicity, simulating real-life human exposure to Cd. Specifically, the study will assess the effect of SSE on sperm count, motility, viability, morphology, DNA integrity, blood testosterone levels, antioxidant capacity, nitro-oxidative stress markers, and the histological structure of testicular tissue.

2. Materials and Methods

2.1. Animals. Sixty adult male mice were sourced from the Pasteur Institute (Tehran, Iran). After a 1-week acclimatization period, the mice were randomly allocated to either the control or experimental groups. The animals were kept under controlled environmental conditions, with a 12-h light/dark cycle, a temperature of $23 \pm 1^\circ\text{C}$, and 50% relative humidity. They had unrestricted access to food (standard diet) and water for the entire study period. All experimental procedures were carried out in compliance with international guidelines for laboratory animal care and were approved by the Animal Welfare and Ethics Committee of Kermanshah University of Medical Sciences, Iran (IR.KUMS.AEC.1401.006).

2.2. Preparation of Plant Extract. The aboveground section of the plant was sourced from the slopes of Zagros in Ilam province. Following approval by expert professors from the Faculty of Pharmacy at Kermanshah University of Medical Sciences, the plant underwent drying in the shade and subsequent grinding into powder using a mill. A quantity of 100 g of the powdered plant material was then subjected to extraction using the percolation method with 90-degree ethanol solvent. Subsequently, the solvent was eliminated using a vacuum distillation apparatus, resulting in the isolation of 10 g of dry extract for our experiment.

2.3. Experimental Design. The animals were allocated into six groups, with each group consisting of 10 mice:

Group 1 (normal control group): As the negative control, which was injected with vehicle.

Group 2 (Cd group): As the positive control group, mice were administered with Cd chloride (CdCl_2) 0.35 mg/kg/day at 10 a.m. by intraperitoneal (i.p.) injections for 35 days.

Group 3 (100 SSE): Mice were treated with 100 mg/kg/day of SSE at 9 a.m. by i.p. injections for 42 days (SSE administration commenced 7 days prior to the onset of CdCl_2 injection and continued throughout the 35-day period of CdCl_2 administration).

Group 4 (200 SSE): Mice were treated with 200 mg/kg/day of SSE at 9 a.m. by i.p. injections for 42 days (SSE administration commenced 7 days prior to the onset of CdCl_2 injection and continued throughout the 35-day period of CdCl_2 administration).

Group 5 (Cd + 100 SSE): Mice received 0.35 mg/kg/day of CdCl_2 and received 100 mg/kg/day of SSE.

Group 6 (Cd + 200 SSE): Mice received 0.35 mg/kg/day of CdCl_2 and received 200 mg/kg/day of SSE.

The doses were selected based on previous studies and a preliminary pilot study [35, 38]. Upon completion of the treatment period, the animals were euthanized, and their testes and epididymides were harvested via laparotomy for sperm analysis. One testis was preserved in fixative for histopathological examination, while the other was immediately frozen for the assessment of oxidant/antioxidant levels [39]. All chemicals used in the study were obtained from Sigma-Aldrich, USA, unless stated otherwise.

2.4. Sperm Parameter Analysis. The epididymis was carefully detached from the testis, and the cauda epididymis was placed in 1 mL of prewarmed Hams F10 medium. The sample was then incubated in a CO_2 incubator for 15 min to allow spermatozoa to swim out. Sperm count was determined using a Neubauer chamber and light microscopy (Olympus Co., Tokyo, Japan), with results expressed as 10^6 cells/mL. Sperm motility was assessed by counting both motile and immotile sperm. Sperm viability was evaluated using Eosin staining, counting 200 sperm per mouse. Morphological abnormalities were examined using Papanicolaou staining and light microscopy at $\times 1000$ magnification. Sperm with head and tail defects were identified and recorded as a percentage of the total sperm count. Furthermore, aniline blue (AB) and toluidine blue (TB) staining and evaluation were performed as described by Moradi et al. [40, 41].

2.5. Histopathological Examination. For histopathological analysis, a single tissue section (thickness = 0.05 μm) was obtained from each testis and examined for any potential alterations in the structure of the germinal epithelium and interstitial connective tissue following the administration of Cd and SSE ($n = 6$) [39].

2.6. Evaluation of Serum Testosterone Levels. Serum testosterone levels were quantified using rat testosterone ELISA kits (Moonblind Inc., USA) through the ELISA method, following the manufacturer's protocol [42, 43].

2.7. Evaluation of Nitro-Oxidative Stress in Testicular Tissue

2.7.1. Measurement of Total Antioxidant Capacity (TAC) in Testicular Tissue. The TAC levels in testicular tissue were assessed using spectrophotometric analysis with a colorimetric assay kit (Naxifer, Navand Salamat Co., Iran) to measure ferric-reducing antioxidant power (FRAP). This method is based on the reduction of ferric ions (Fe^{3+}) to ferrous ions (Fe^{2+}) in the presence of 2,4,6-tripyridyl-S-triazine (TPTZ), resulting in a blue-colored complex with maximum absorbance at 593 nm. The FRAP values were expressed as nanomoles of Fe^{2+} equivalents per milligram of wet tissue weight (nmol/mg protein) [39, 44].

2.7.2. Evaluation of Lipid Peroxidation (LPO) in Testicular Tissue. LPO levels in testicular tissue samples were assessed by measuring malondialdehyde (MDA) using the TBA reaction with a Nalondi assay kit (Navand Salamat Co., Iran). Briefly, 0.2 mL of tissue homogenate was mixed with 0.8 mL of working solution containing phosphoric acid and TBA, then added to the samples. The mixture was heated at 90°C for 45 min and subsequently cooled on ice. The thiobarbituric acid-reactive substances (TBARS) were quantified by measuring absorbance at 532 nm, based on an MDA standard curve. TBARS levels were expressed as MDA equivalents per milligram of wet tissue weight (nmol/mg protein) [39, 45].

2.7.3. Assessment of Nitric Oxide (NO) Levels in Testicular Tissue. The total NO content in homogenized testicular tissue was measured using the Griess reaction, following the manufacturer's instructions (Natrix assay kit; Navand Salamat Co., Iran). In this reaction, NO is rapidly converted to nitrite in an acidic environment, then to HNO_2 . Sulfanilamide is added, enabling HNO_2 to form a diazonium salt, which subsequently reacts with N-(1-naphthyl) ethylenediamine dihydrochloride to produce an azo dye measurable at 570 nm. NO levels in the samples were reported as micromoles per milligram of protein ($\mu\text{mol/mg protein}$) [42, 44].

2.8. Statistical Analysis. Data analysis was performed using SPSS software (version 26). The normality and homogeneity of the data were assessed using the Kolmogorov-Smirnov test. Quantitative data are presented as mean $\pm$ standard deviation and 95% confidence intervals (CI). Differences between experimental groups were analyzed using one-way ANOVA, followed by Dunnett's post hoc test for pairwise comparisons. The LSD post hoc test was also used for comparisons among the intervention groups. Statistical significance was set at $p \leq 0.05$.

3. Results

3.1. S. striata Alleviates the Cd-Induced Reduction in Testicular Weight. After treatment, the testis weight was significantly lower in the Cd and Cd + SSE groups compared to the control group ($p < 0.05$), suggesting Cd-induced testicular toxicity. Although SSE effectively helped maintain testis weight, a significant difference remained between the Cd + SSE groups and the control group ($p < 0.05$) (Figure 1).

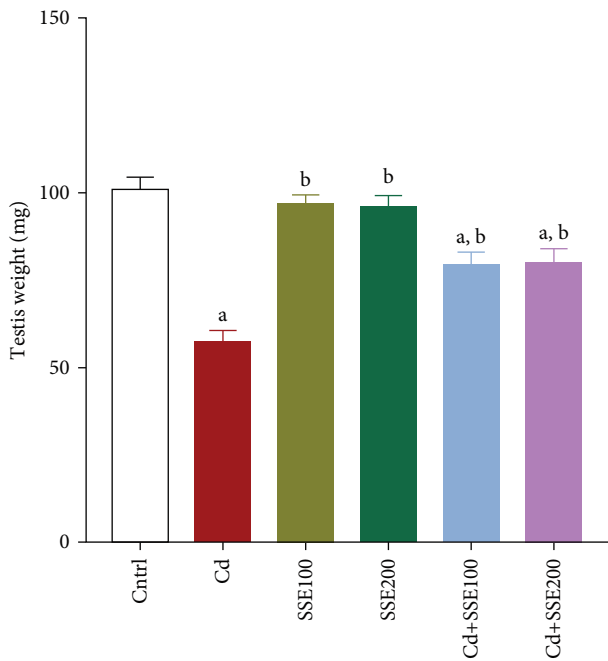


FIGURE 1: The effect of Cd with or without SSE on testis weight (mg). Cntrl: control; SSE100 and SSE 200: mice injected with 100 or 200 mg/kg of SSE, Cd: mice treated with only Cd; Cd + SSE100 and Cd + SSE200: mice treated with Cd plus 100 and 200 mg/kg of SSE, respectively. ^a $p < 0.05$ vs. control group; ^b $p < 0.05$ vs. Cd group. Cd, cadmium; SSE, *Scrophularia striata* extract.

3.2. *S. striata* Reverses the Cd-Induced Damage to Spermatogenesis. Histopathological examination of the control groups showed normal seminiferous tubule architecture, with a well-organized arrangement of germ cells at different stages of development and lumens filled with spermatozoa (Figure 2A). Likewise, the groups treated with SSE alone exhibited normal histological features in the seminiferous tubules, with a structured alignment of germ cells throughout spermatogenesis (Figure 2B,C). In contrast, the Cd-treated group displayed significant degenerative changes in the seminiferous tubule epithelium, including vacuole formation, separation of the germinal epithelium, necrosis, interstitial edema, and disrupted lumens, indicating substantial Cd-induced testicular damage (Figure 2D). SSE treatment markedly alleviated the histological damage induced by Cd exposure, with all SSE doses improving the testicular architecture (Figure 2E,F). Collectively, these findings suggest that Cd exposure triggered apoptosis in spermatogenic cells, halting spermatogenesis, an effect that was significantly reversed by SSE administration (Figure 2).

3.3. *S. striata* Mitigates the Changes in Testosterone Levels Induced by Cd Exposure. The Cd-treated group showed a significant decrease in testosterone levels compared to the control group ($p < 0.05$). In contrast, treatment with 100 and 200 mg/kg of SSE significantly restored serum testosterone levels ($p < 0.05$), suggesting a potential protective effect of SSE in maintaining the function of testicular cells (Figure 3).

3.4. *S. striata* Reduces Cd-Induced Disturbances in Sperm Parameters. Cd treatment resulted in a significant decrease in sperm count ($p < 0.05$), but treatment with SSE facilitated recovery in the Cd-exposed mice ($p < 0.05$) (Figure 4A). Additionally, sperm motility was significantly reduced in the Cd-treated group compared to controls ($p < 0.05$), whereas SSE treatment significantly improved motility in both Cd + SSE groups ($p < 0.05$) (Figure 4B). The Cd-exposed group also exhibited a notable increase in the number of dead sperm compared to controls ($p < 0.05$), while SSE treatment significantly enhanced sperm viability in comparison to the Cd group ($p < 0.05$) (Figure 4C). Furthermore, Cd exposure led to a significant increase in the percentage of abnormal sperm morphology compared to the control group ($p < 0.05$). SSE therapy reduced these morphological abnormalities, although a significant difference remained between the Cd + SSE groups and the control group ($p < 0.05$) (Figure 4D).

3.5. *S. striata* Alleviates Cd-Induced Impairment in Sperm DNA Integrity. Evaluation of sperm chromatin quality revealed a significant rise in the AB and TB index in the Cd group by comparison to the control group ($p < 0.05$). In contrast, SSE therapy significantly decreased the proportion of AB+ sperm relative to the Cd group ($p < 0.05$) and substantially reduced the proportion of TB+ sperm compared to the Cd-exposed group ($p < 0.05$) (Figure 5A,B).

3.6. *S. striata* Modulates Testicular Nitro-Oxidative Status Following Cd Exposure. MDA levels in the testes were significantly higher in Cd-treated animals compared to the control group ($p < 0.05$), indicating oxidative stress induced by Cd exposure. In contrast, treatment with 100 and 200 mg/kg of SSE significantly reduced MDA levels ($p < 0.05$) (Figure 6A). Regarding NO levels, a marked difference was observed between the Cd-treated and control groups ($p < 0.05$). However, supplementation with SSE in the Cd + SSE groups led to a significant reduction in NO levels ($p < 0.05$), suggesting that SSE effectively alleviated nitrosative stress in the testes (Figure 6B). Furthermore, TAC levels were significantly lower in the Cd-treated group compared to the control group ($p < 0.05$). Importantly, the TAC levels in the SSE-treated groups were significantly higher than those in the Cd-treated group ($p < 0.05$), indicating a restoration of the antioxidant status in the testes. Nonetheless, a significant difference between the Cd + SSE groups and the control group persisted ($p < 0.05$) (Figure 6C).

4. Discussion

Over the past five decades, the decline in male fertility has emerged as a notable public health concern, primarily because of widespread exposure to environmental toxins globally. Research indicates that Cd represents a substantial hazard to both the environment and human health, which environmental and occupational exposure to this toxic pollutant have detrimental effects on human male fertility [46–48]. *S. striata* is a traditional herbal remedy known for its promising antioxidant and anti-inflammatory properties. To the best of our knowledge, for the first time, we deciphered that SSE protects

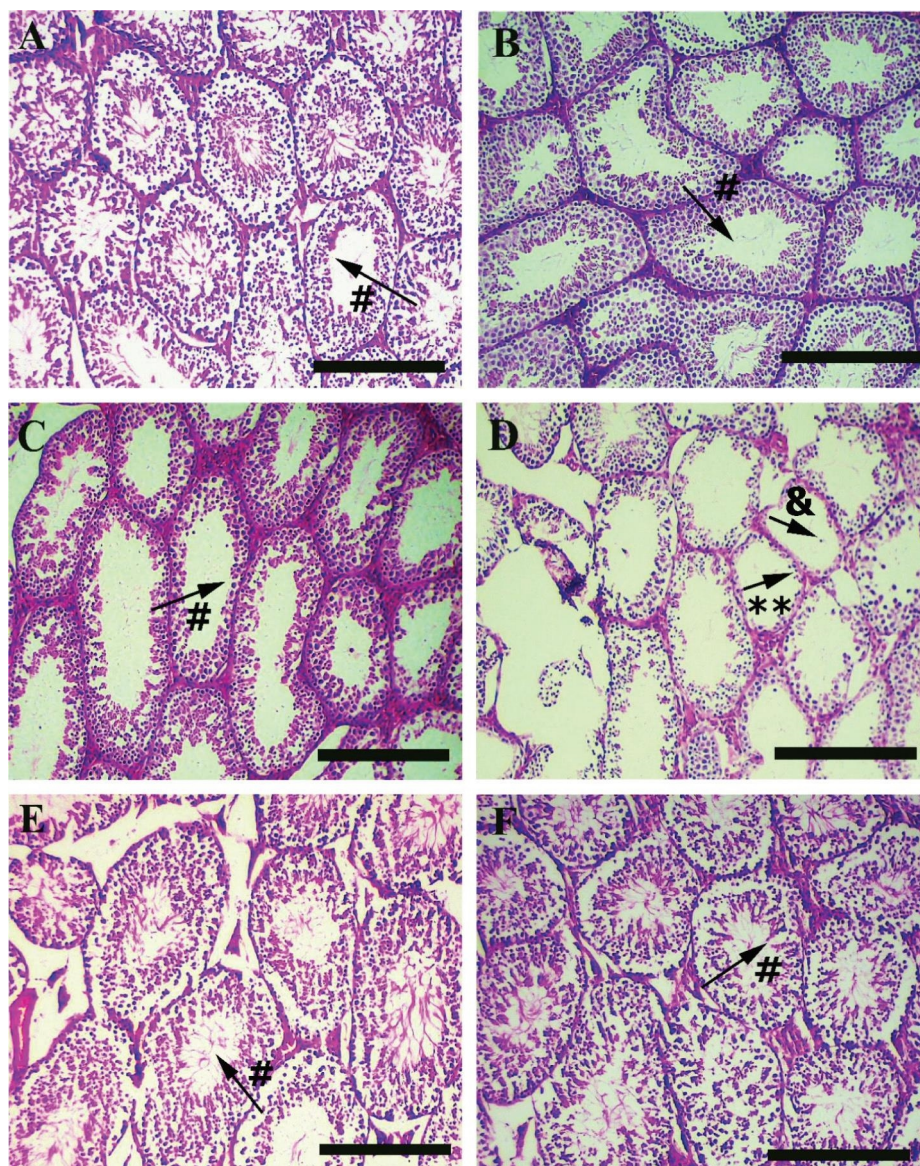


FIGURE 2: Light microscope structure of the testis in the controls and experimental groups. (A, B, and C) testicular tissue in the control group and the control groups were administered with 100 and 200 mg/kg of SSE, respectively, indicating the normal structure of the testis and the germinal epithelium. #: The cellular layers within the seminiferous tubules are well-organized and complete. (D) Damaged testicular tissue in the Cd-treated mice with degenerative changes. **: The cell layers in the seminiferous tubules are incomplete. &: Necrotic cells are observed within the seminiferous tubule lumen. (E and F) Testis structure in the Cd + SSE100 and Cd + SSE200 groups, respectively, revealing improvement of testicular tissue toward normal appearance (H&E staining, 100x, scale bar = 100 μ m, n = 6). Cd, cadmium; SSE, *Scrophularia striata* extract.

against Cd-induced male reproductive toxicity by improving antioxidative pathways in the testis.

Histopathological analysis revealed that SSE treatment helped maintain the normal histological and morphological integrity of the testes in mice exposed to Cd. Cd administration caused damage to Leydig and Sertoli cells, likely due to the induction of oxidative stress [6, 7, 49]. Previous research suggested that Cd treatment could deteriorate Sertoli and/or Leydig cells, affecting testosterone levels and testicular necrosis and atrophy [9, 28, 50]. These effects may be explained by Cd's ability to disrupt the integrity of the blood–testes barrier, induce oxidative stress and inflammatory cytokines like

TNF- α , cause DNA damage, and trigger germ cell apoptosis [6, 9, 12]. Spermatogenesis is a multifaceted process influenced by genetic, hormonal, and environmental factors, and dysfunction in any of these aspects results in lower sperm quality, ultimately leading to infertility [43, 51, 52]. SSE treatment, however, restored the structure of seminiferous tubules and various stages of spermatogenesis, probably thanks to the modulation of oxidative stress by its antioxidant components, such as quercetin and phenolic compounds found in this herbal remedy [35, 36].

In terms of hormone levels, a noteworthy reduction in plasma testosterone was noted in the Cd-treated group,

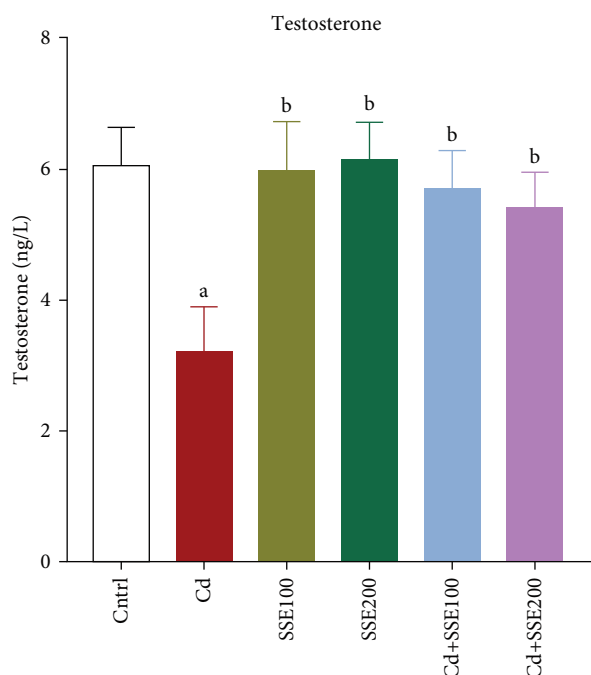


FIGURE 3: The effect of Cd with or without SSE on testosterone levels in the controls and experimental groups. Cntrl: control; SSE100 and SSE 200: mice injected with 100 or 200 mg/kg of SSE, Cd: mice treated with only Cd; Cd + SSE100 and Cd + SSE200: mice treated with Cd plus 100 and 200 mg/kg of SSE, respectively. ^a $p < 0.05$ vs. control group; ^b $p < 0.05$ vs. Cd group. Cd, cadmium; SSE, *Scrophularia striata* extract.

consistent with research suggesting that Cd exposure leads to disruptions in steroid biosynthesis and serum testosterone levels [3, 28]. Indeed, Cd disrupts hormone production through direct binding to androgenic and estrogenic receptors, altering the expression of steroidogenesis enzymes and inhibiting LH receptor expression [3, 29, 49]. In addition, Cd indirectly impacts hormone secretion by hindering noradrenaline release, altering GnRH release, and impacting LH and prolactin secretion [53]. In the current study, the impairment in sperm parameters could be attributed to compromised spermatogenesis linked with decreased testosterone synthesis in the testicular tissue [42]. It is striking that 100 and 200 mg/kg of SSE could remarkably diminish the negative impact of Cd on testosterone production, which can be linked to the ability of the potent antioxidant properties of SSE extract components to regulate the nitro-oxidative stress in the testis [36]. In other words, these enhancements could be attributed to the extract's antioxidant properties, which protect Leydig cells from damage caused by free radicals and inhibit steroidogenesis in the testes [45, 54]. Our results align with previous research indicating that flavonoids found in herbal extracts have the ability to increase testosterone levels, thereby enhancing sperm quality [16, 55].

The injection of Cd induced a notable decrease in sperm count, viability, and motility, accompanied by a notable increase in abnormal morphology. These findings mirror previous human and animal studies indicating that Cd induces a decrease in the sperm concentration, motility, and viability while increasing the number of sperm with abnormal morphology

[28, 50, 56, 57]. These findings could be attributed to Cd-induced oxidative stress and apoptosis pathways, which results in a subsequent decline in semen quality [19, 27, 49, 56]. Moreover, these findings align with previous studies indicating that even low levels of Cd in semen may potentially lead to a deterioration of sperm quality by inducing oxidative damage [6, 46, 49]. More importantly, Cd treatment notably elevated the AB and TB indexes in spermatozoa, indicating chromatin fragmentation in sperm. Similarly, some studies found that men with idiopathic infertility had heightened Cd levels in their semen, which correlated with markedly reduced sperm DNA integrity and higher levels of ROS in the semen [1, 13, 58]. It has also been demonstrated that Cd causes irregular histone modifications in the sperm nucleus during spermiogenesis, leading to damage in the sperm's nuclear DNA [1, 13, 58].

Conversely, SSE treatment effectively protects sperm quality and chromatin integrity from the spermatotoxic effects induced by Cd exposure. Although no research has directly investigated the impact of SSE on the reproductive system, we attribute these positive outcomes to its capacity to neutralize free radicals, either by inhibiting their formation or by scavenging and eliminating already existing reactive species [35]. With constituents like quercetin and phenolic compounds, which are responsible for antioxidant characteristics, SSE is likely to maintain mitochondrial function by scavenging ROS, thus promoting spermatogenesis and improving sperm motility, vitality, and morphology [35, 39, 59]. An alternative explanation for the enhancement of sperm parameters in SSE-treated mice could be related to testosterone, as we found a significant increase in testosterone levels following treatment with this extract [59, 60]. Thus, SSE attenuates male reproductive toxicity and sperm dysfunction probably through its antioxidant and testosterone-boosting properties, which enhance testicular function [39].

The findings revealed a notable reduction in TAC alongside a notable increase in NO and MDA levels, representing that Cd treatment disrupts the balance in the redox system, leading to significant nitro-oxidative stress in the testes of mice. Recent studies have reported that Cd can trigger oxidative stress and LPO across a broad range of organs and tissues [6, 12, 28]. Free active radicals and oxidative stress are central to the development and progression of male infertility and testicular dysfunction caused by Cd exposure [11, 28, 47, 50]. Cd-induced testicular oxidative stress is driven by the production of ROS, a reduction in glutathione (GSH) levels, elevated LPO, and the inhibition of antioxidant enzymes such as catalase and superoxide dismutase (SOD). Additionally, Cd interacts with SH groups in membrane and intracellular proteins, further disrupting cellular function [11, 12, 54, 61].

As a natural antioxidant, SSE played a crucial role in regulating and averting nitro-oxidative injury resulting from Cd exposure via inhibition of NO and MDA generation while also enhancing the total antioxidant defense mechanisms in the testes. These observations are in line with a study by Amri et al. [62], where the authors noted significant modulation of hepatic myeloperoxidase (MPO), glutathione

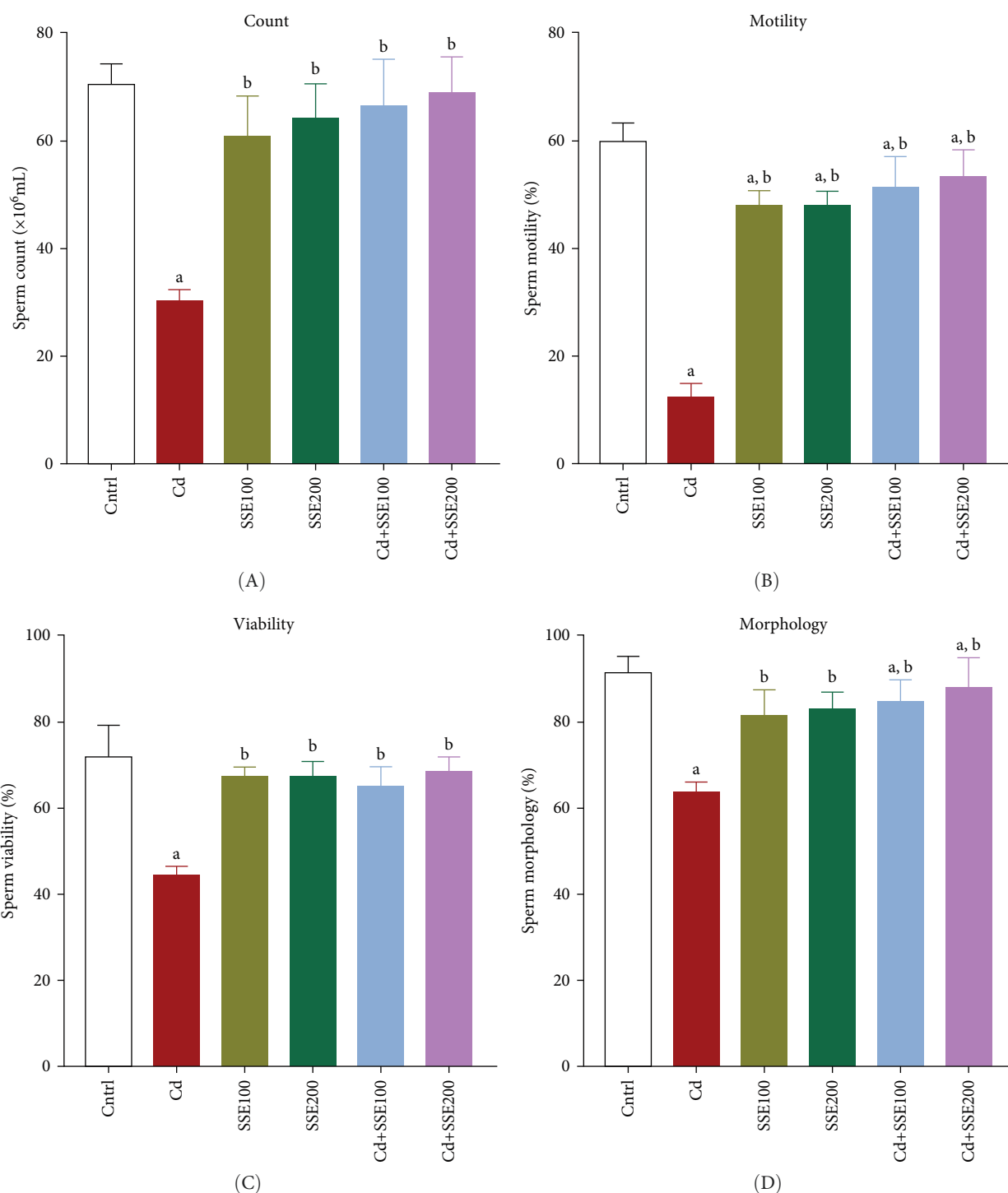


FIGURE 4: The effect of Cd with or without SSE on sperm count (A), sperm motility (B), sperm viability (C), and sperm morphology (D) in the control and experimental groups. Cntrl: control; SSE100 and SSE200: mice injected with 100 or 200 mg/kg of SSE, Cd: mice treated with only Cd; Cd + SSE100 and Cd + SSE200: mice treated with Cd plus 100 and 200 mg/kg of SSE, respectively. ^a $p < 0.05$ vs. control group; ^b $p < 0.05$ vs. Cd group. Cd, cadmium; SSE, *Scrophularia striata* extract.

peroxidase (GPX), MDA, TAC, and SOD levels in cirrhotic rats following treatment with the hydroalcoholic extract of SSE [62, 63]. Recent studies have revealed that *S. striata* contain compounds like flavonoids, cinnamic acid, phenylpropanoids, neptrin, flavonoid glycoside, acteoside, quercetin, and

phenylpropanoid glycoside, many of which exhibit strong antioxidant properties [33, 35, 36]. This unique complex of bioactive components has the ability to restore the oxidant/antioxidant balance in testicular tissue, consequently protecting spermatogenesis from oxidative damage induced by Cd.

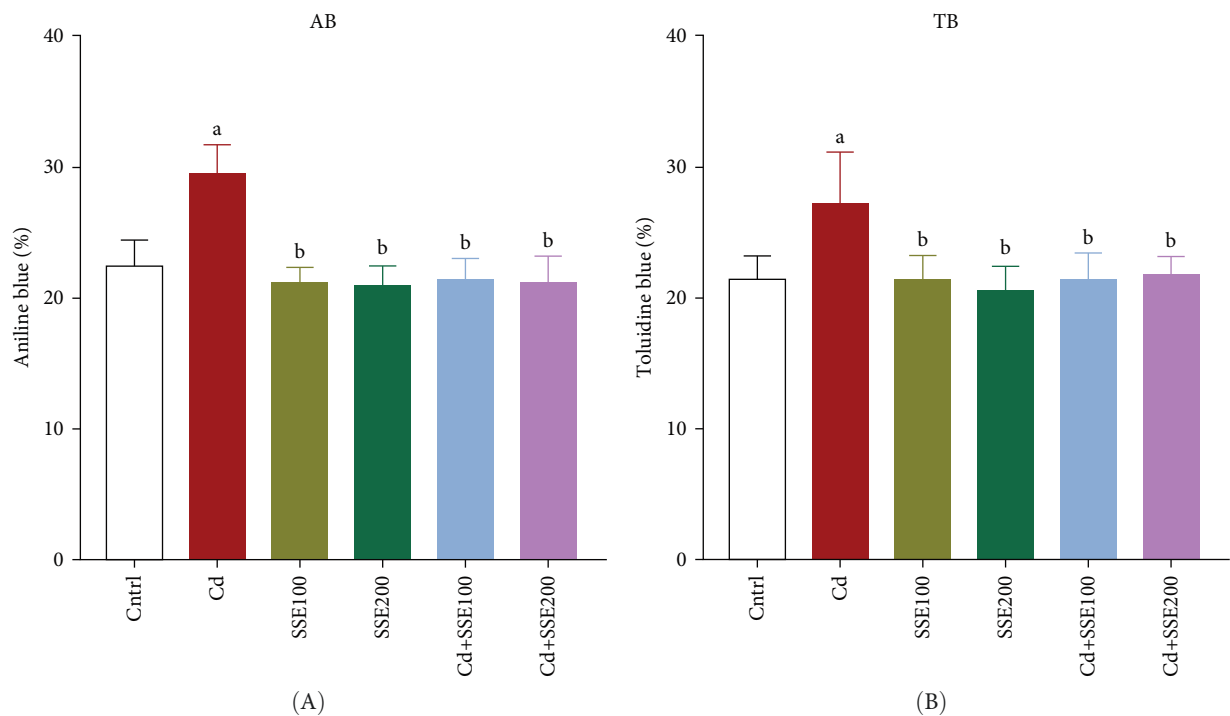


FIGURE 5: The effect of Cd with or without SSE on the percentage of AB+ (A) and TB+ (B) sperm in the control and experimental groups. Cntrl: control; SSE100 and SSE200: mice injected with 100 or 200 mg/kg of SSE, Cd: mice treated with only Cd; Cd + SSE100 and Cd+SSE200: mice treated with Cd plus 100 and 200 mg/kg of SSE, respectively. ^a $p < 0.05$ vs. control group; ^b $p < 0.05$ vs. Cd group. Cd, cadmium; SSE, *Scrophularia striata* extract.

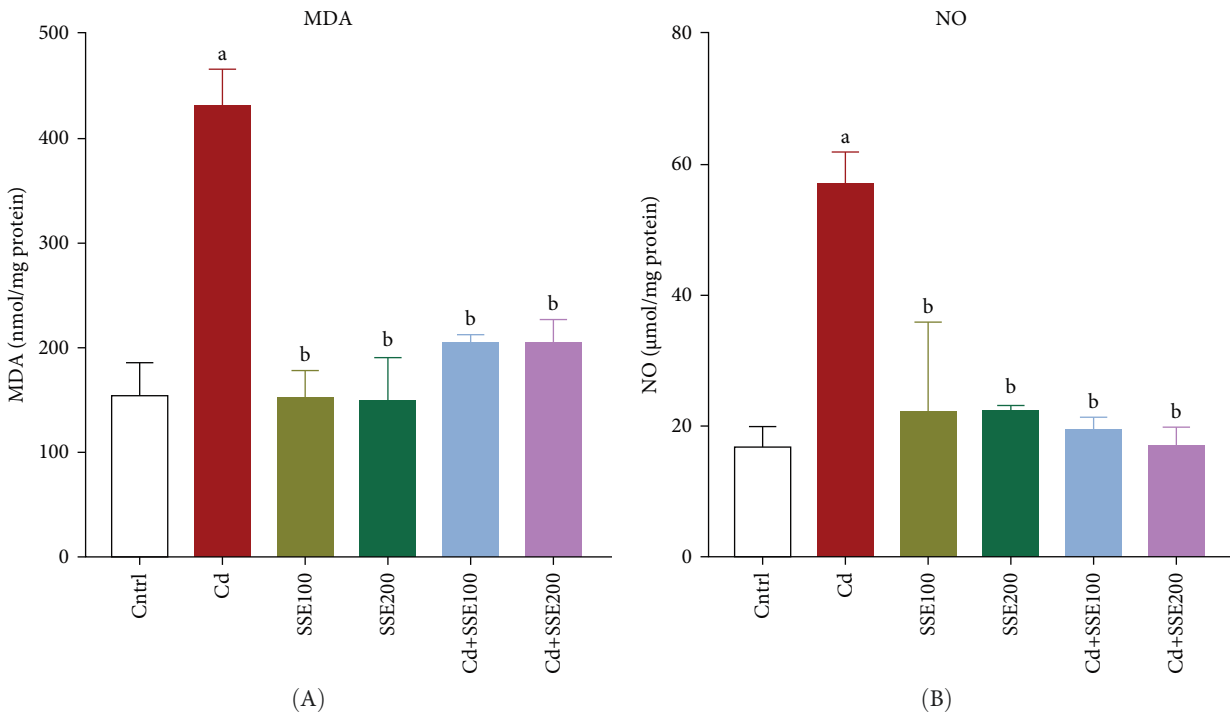


FIGURE 6: Continued.

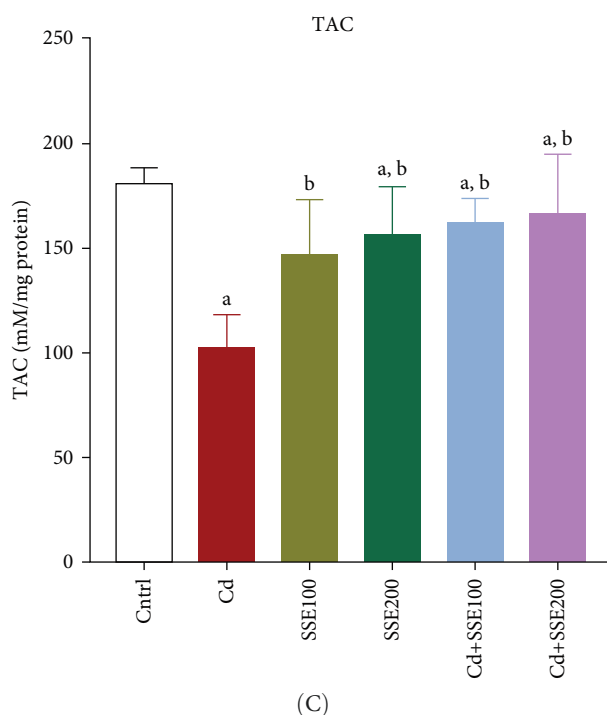


FIGURE 6: The effect of Cd with or without SSE on MDA (A), NO (B), and TAC (C). Cntrl: control; SSE100 and SSE200: mice injected with 100 or 200 mg/kg of SSE, Cd: mice treated with only Cd; Cd + SSE100 and Cd+SSE200: mice treated with Cd plus 100 and 200 mg/kg of SSE, respectively. ^a $p < 0.05$ vs. control group; ^b $p < 0.05$ vs. Cd group. Cd, cadmium; MDA, malondialdehyde; NO, nitric oxide; SSE, *Scrophularia striata* extract; TAC, total antioxidant capacity.

In fact, the phenolic compounds present in plant-derived extracts exhibit remarkable antioxidant properties due to their structural features and the hydroxyl groups in their aromatic rings. Acting as hydrogen donors, these compounds can effectively mitigate inflammation and apoptosis in the testes, thereby supporting enhanced testicular function [36, 45]. It is worth noting that this is the first report showing that SSE enhances antioxidant defenses in the testes by scavenging free radicals, eliminating oxidative agents, and increasing overall antioxidant capacity.

5. Conclusions

The extract of *S. striata* demonstrate significant protective effects on testes' function and sperm quality against Cd-induced reproductive toxicity. These benefits are attributed to the extract's potent antioxidant properties, particularly its phenolic and flavonoid compounds that act as effective free radical scavengers. Nevertheless, further in-depth studies are required to uncover and characterize the specific bioactive constituents present in the extract of *S. striata*. This research will facilitate the development of novel herbal remedies to combat reproductive toxicity and male sub/infertility associated with environmental pollutants.

Data Availability Statement

The data that support the findings of this study are available on request from the corresponding author. The data are not publicly available due to privacy or ethical restrictions.

Disclosure

The authors declare that all data were generated in-house and that no paper mill was used. This work was performed in partial fulfillment of the requirements for the Medical degree of Saman Mirzaei in the School of Medicine, Kermanshah University of Medical Sciences, Kermanshah, Iran.

Conflicts of Interest

The authors declare no conflicts of interest.

Author Contributions

Mojtaba Moradi, Azita Faramarzi, and Saman Mirzaei conceived the study and designed the research plan. Saman Moradi and Saman Mirzaei conducted animal experimentation. Elham Ghanbari and Fereshteh Afkar designed and conducted the histological analysis. Mojtaba Moradi, Azita Faramarzi, and Saman Mirzaei conducted sperm analysis. Leila Rezakhani, Mojtaba Moradi, Masoumeh Golestan Jahromi, and Saman Moradi designed and conducted the biochemical and molecular analysis. Leila Rezakhani and Azita Faramarzi analyzed the data. Mojtaba Moradi wrote the manuscript draft. Mojtaba Moradi and Azita Faramarzi edited the manuscript. All authors read and approved the final manuscript.

Funding

This work was supported by the Kermanshah University of Medical Sciences, Iran (grant number IR.KUMS.AEC.1401.006).

Azita Faramarzi has received research support from Kermanshah University of Medical Sciences.

Acknowledgments

The authors would like to express their gratitude to the staff of the Fertility and Infertility Research Center, Kermanshah, Iran. Additionally, we are grateful to Prof. Dr. Mozafar Khazaei, Prof. Dr. Mitra Bakhtiari, Dr. Mohsen Shayestehekta, and Bahareh Moradi for their technical assistance.

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